SOFTWARE REQUIREMENTS AND SPECIFICATIONS

In software engineering, every successful software project begins with a clear understanding of what the system should do. This understanding is achieved through two critical concepts **software requirements and software specifications.** Requirements define the needs and expectations of users, while specifications describe these needs in a structured and detailed form that guides developers. Without well-defined requirements and specifications, software development risks failure, misunderstanding, and costly rework. Software requirements and specifications are the foundation of every software project. They transform vague ideas into structured, measurable objectives that guide development, testing, and maintenance. A well-prepared SRS ensures that both developers and clients share a common understanding, reducing risks, improving quality, and ensuring that the software meets real-world needs. In essence, the success of any software system depends largely on the clarity and completeness of its requirements and specifications.

Software Requirements

A software requirement is a detailed description of a service, function, or constraint that a system must provide or satisfy. It answers the question: What should the software do? Requirements are gathered from **stakeholders such as end-users, clients, managers, and regulatory bodies.** A software product that fails to meet its requirements is considered unsuccessful, regardless of its technical excellence.

Types of Requirements:

Functional Requirements – Define specific actions or behaviors the system should perform. Example: "The system shall allow users to log in using their email and password."

Non-Functional Requirements – Define how the system should perform. Example: "The system shall respond to user queries within two seconds." These include performance, usability, reliability, and security requirements.

Importance of Software Requirements

1. They serve as the foundation for software design and development.
2. They ensure clear communication between clients and developers.
3. They act as a reference for testing and validation of the final product.
4. They help in project estimation, planning, and risk management.

Software Specification

After collecting and analyzing requirements, they are documented in a formal way known as the Software Requirements Specification (SRS). The SRS provides a precise and unambiguous description of the software system to be built. It serves as a contract between the customer and the development team.

Characteristics of a Good SRS

A good Software Requirements Specification should be:

1. Correct – Accurately reflects stakeholder needs.
2. Complete – Contains all required functionalities.

3. Unambiguous – Clear and precise with no room for multiple interpretations.
4. Verifiable – Each requirement can be tested or validated.
5. Modifiable – Easy to update when changes occur.
6. Traceable – Each requirement can be linked to its source and corresponding design or test case.

Relationship between Requirements and Specifications

1. Requirements expresses user goals in natural language while Specification converts those goals into technical documentation.
2. Requirements are user-focused, specifications are developer-focused
3. The requirements represent what users need, while the specification defines how those needs are formally described. Example:

Requirement: “The user should be able to print receipts.”

Specification: “Upon completing a transaction, the system shall send a print command to the connected printer to produce a receipt.” Thus, the specification converts user intentions into technical instructions for developers.